SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I N Naveenkumar (192521412) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *N Naveenkumar*

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

ARTICLE REVIEW – ASSESSMENT

Deep Reinforcement Learning for Personalized Treatment Recommendation

Selected Research Article

This review selects a peer-reviewed, open-access research article published in 2022 in Statistics in Medicine. The article directly matches the syllabus topic of Reinforcement Learning Applications and focuses on personalized treatment recommendation in precision medicine.

Article Citation Details

Liu, M., Shen, X., & Pan, W. (2022). Deep reinforcement learning for personalized treatment recommendation. Statistics in Medicine, 41(20), 4034–4056. DOI: 10.1002/sim.9491.

Valid DOI link: https://doi.org/10.1002/sim.9491

Publisher: Wiley Online Library. The article is open access.

Task 1 – Article Summary

1(a) Domain and Specific ML Problem

The article belongs to precision medicine, cancer drug response prediction, personalized treatment recommendation, and deep reinforcement learning. The central machine-learning problem is not simply predicting the response to one drug at a time. Instead, the authors aim to learn a policy that can rank candidate drugs for a particular cancer cell line or patient according to expected treatment sensitivity. This is formulated as a sequential decision-making problem using a Markov decision process. The work introduces Proximal Policy Optimization Ranking (PPORank), a deep reinforcement learning approach that directly optimizes the ranking quality of recommended treatments.

1(b) Key Objective and Research Gap (150–200 words)

The article addresses the problem of selecting and ranking personalized cancer treatments from high-dimensional molecular information. Existing approaches mainly use supervised learning to predict a numerical drug-response value, such as drug sensitivity, and then use those predictions to select treatments. The authors identify a gap: clinical treatment selection is sequential, and in practice the most important output is often a short ranking of the best treatment options rather than an accurate prediction for every individual drug. A conventional supervised model also does not naturally represent delayed outcomes or learning as new observations become available. To address this gap, the authors propose PPORank, a deep reinforcement learning method based on Proximal Policy Optimization and an actor–critic framework. The method treats drug ranking as a sequential decision process and uses NDCG as a long-term reward, allowing the learning objective to directly match the desired ranking quality. Experiments on large cancer cell-line datasets and simulated data show that PPORank generally outperforms the compared supervised and ranking methods. The study therefore demonstrates the potential of reinforcement learning for personalized treatment recommendation while recognizing that further validation is required before clinical deployment.

Task 2 – Methodology Analysis

2(a) ML Technique and Dataset

The main technique is Proximal Policy Optimization Ranking (PPORank), a deep reinforcement learning approach. The policy is parameterized using a Deep & Cross Network, while a GRU component is used to incorporate dynamic evaluation information. The method follows an actor–critic structure and uses PPO optimization. Instead of using a simple prediction loss, the method directly optimizes NDCG, a ranking-quality metric.

The main cancer drug-screening data sources are the Genomics of Drug Sensitivity in Cancer (GDSC) and Cancer Cell Line Encyclopedia (CCLE). GDSC contains 1,001 cancer cell lines across 31 cancer types and 265 pharmacological compounds in the described dataset. The cell lines include gene-expression, whole-exome sequencing, copy-number variation and DNA-methylation information. After preprocessing and removing toxic drugs, the study uses 223 drugs for GDSC and 19 drugs for CCLE. Drug response is represented using log(IC50)-based sensitivity information, with drug-specific thresholds used to normalize sensitivity scores.

For external validation, the authors use the TCGA breast-cancer cohort. They identify HER2-positive, triple-negative breast cancer and BRCA1/2-mutant patient groups and harmonize gene-expression information between TCGA and GDSC. The study also includes simulated datasets to evaluate behavior under linear and nonlinear relationships.

2(b) Mapping the Technique to CO4 Topics

CO4 Topic	Connection to the Article	Evidence / Explanation
Inductive Learning	Learning a policy from observed data	The model learns treatment-ranking behavior from cell-line and drug-response observations.
Decision Trees	Not the main technique	Decision Trees are not used as the core model; they are relevant as an interpretable alternative for supervised classification.
Statistical Learning	Baseline and comparison methods	The paper compares PPORank with supervised approaches including elastic net, kernel regression and other ranking/recommendation methods.
Reinforcement Learning	Primary CO4 topic	PPORank models ranking as sequential decision-making and uses PPO, actor–critic learning and NDCG-based rewards.

Methodological Strength

A major strength is that the learning objective is closely aligned with the actual recommendation task. Instead of optimizing only drug-response prediction error, PPORank directly optimizes NDCG, which measures the quality of the ranked treatment list. The authors also test sequential learning by starting with part of the GDSC data and adding new cell lines over time, making the experiment more representative of continuous data collection.

Methodological Limitation

The most important limitation is that much of the evidence is based on retrospective cell-line screening data and simulation rather than prospective clinical treatment outcomes. A cell line is not a complete representation of a patient, and drug sensitivity in vitro may not translate directly into clinical effectiveness, toxicity or patient preference. The complexity of the deep RL model also makes independent validation and clinical interpretability more challenging.

Task 3 – Results & Critical Evaluation

3(a) Key Results

The main evaluation metric is NDCG rather than accuracy, because the task is ranking candidate treatments. On the full CCLE dataset, PPORank achieved a mean NDCG of 0.7611, compared with 0.7432 for CaDRRes and 0.7468 for elastic net. On GDSC, PPORank showed a larger and more consistent improvement over competing methods and had the smallest reported standard deviation (0.003), while CaDRRes had a standard deviation of 0.011.

In the primary simulation experiments, PPORank achieved NDCG values of 0.832 (SD 0.04) for Scenario 1 with 10,000 samples, 0.705 (SD 0.05) for Scenario 2 with 10,000 samples, and 0.959 (SD 0.03) for Scenario 3 with 10,000 samples. Scenario 3 represented a highly nonlinear setting, where PPORank showed particularly strong performance.

Experiment	PPORank	Best/Key Comparator	Interpretation
CCLE, full dataset	Mean NDCG = 0.7611	CaDRRes = 0.7432	Moderate improvement
GDSC, full dataset	Best overall; SD = 0.003	CaDRRes SD = 0.011	More consistent improvement
Simulation 1, n=10,000	0.832 ± 0.04	CaDRRes = 0.824 ± 0.02	PPORank slightly higher
Simulation 2, n=10,000	0.705 ± 0.05	CaDRRes = 0.682 ± 0.02	PPORank higher
Simulation 3, n=10,000	0.959 ± 0.03	CaDRRes = 0.931 ± 0.02	Strong advantage in nonlinear case

3(b) Critical Evaluation

The reported metrics are appropriate for the research question because NDCG evaluates whether the most relevant treatments appear near the top of the recommendation list. The use of five-fold cross-validation for major dataset comparisons and three-fold cross-validation for some top-k analyses provides repeated estimates rather than relying on one split. However, strong retrospective ranking performance should not be interpreted as proof of clinical benefit.

Potential sources of bias include differences between cell-line biology and real patient biology, drug-screening selection effects, the use of imputed or harmonized patient features during external validation, and the possibility that performance varies across cancer subtypes. Additional experiments that would strengthen the findings include prospective clinical validation, independent hospital datasets, subgroup fairness analysis, uncertainty estimation, comparison with clinician-selected treatment rankings, and evaluation of safety and toxicity outcomes.

3(c) Real-World Applicability

A system based on this approach could potentially support oncologists by producing a ranked list of candidate treatments from patient-specific molecular and clinical information. However, it should be treated as a clinical decision-support tool rather than an autonomous prescribing system. Scaling requires access to high-quality longitudinal electronic health records, molecular profiles, treatment histories and reliable outcome labels. Training and evaluating deep reinforcement learning models also require substantial computational resources.

Major practical challenges include privacy and governance of patient data, interoperability between hospitals, distribution shifts across populations, incomplete clinical records, model drift, explainability, fairness, and the need for prospective clinical trials. Ethical deployment requires clinician oversight, patient consent where appropriate, audit trails, safety constraints and clear communication of model uncertainty.

Task 4 – Reflection & Learning Outcome

4(a) Three Key Insights

1. Reinforcement learning is especially useful when decisions occur sequentially and the effect of an action can appear later. This differs from one-step supervised prediction.
2. The evaluation metric should match the real task. PPORank directly optimizes NDCG because the practical goal is to rank the best treatment options, not merely predict an individual response value.
3. High machine-learning performance does not automatically mean clinical readiness. Patient-level validation, safety, fairness, interpretability and prospective evaluation are essential before deployment.

Connection to CO4

The article deepens the CO4 concept of Reinforcement Learning Applications by showing how an abstract Markov decision process can be used to represent a real-world sequential recommendation problem. It also demonstrates how statistical learning and supervised methods can act as baselines for evaluating a reinforcement-learning approach.

4(b) Proposed Novel Extension

A feasible extension would be to evaluate PPORank on a multi-hospital longitudinal electronic health-record dataset that combines molecular information, previous treatment history, toxicity events and patient outcomes. The experiment could compare standard PPORank with a safety-constrained PPORank that applies clinically defined action constraints and uncertainty thresholds before presenting a recommendation.

Expected impact: this experiment would test whether the model remains useful when the data distribution changes across hospitals and patient populations, while also measuring safety-related outcomes. Evaluation could include NDCG@1, NDCG@5, calibration, subgroup performance, uncertainty, treatment-related adverse events and clinician agreement. The experiment is feasible as an offline research study if an appropriately governed dataset is available, but clinical use would still require prospective validation and ethical approval.

Overall Conclusion

Liu, Shen and Pan present PPORank as a promising deep reinforcement learning framework for personalized treatment recommendation. The paper's main contribution is to formulate treatment ranking as sequential decision-making and optimize the ranking objective directly through PPO and NDCG. The experimental results show consistent advantages over several comparison methods, particularly in larger and nonlinear settings. Nevertheless, the evidence remains primarily computational and retrospective. The most important next step is rigorous clinical and safety validation using independent longitudinal patient data. Overall, the article provides a strong example of how reinforcement learning can extend the CO4 concepts from theoretical algorithms to practical precision-medicine applications.

References

1. Liu, M., Shen, X., & Pan, W. (2022). Deep reinforcement learning for personalized treatment recommendation. Statistics in Medicine, 41(20), 4034–4056. https://doi.org/10.1002/sim.9491

2. Yu, C., Liu, J., Nemati, S., & Yin, G. (2022). Reinforcement Learning in Healthcare: A Survey. ACM Computing Surveys, 55(1), Article 5. https://doi.org/10.1145/3477600

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand